



Cellular Automata for Multiple Biome Cave Generation in Video Games

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Introduction

Video games often implement sprawling cave systems into their world generation. A more interactive environment keeps content exciting for users.

This project shows how **Cellular Automata** rules can be utilized in cave generation. While this method has been done before, this project focused on how different rules and parameters could be applied to different “Biomes”, effectively allowing for uniquely generated caves depending on the players’ location

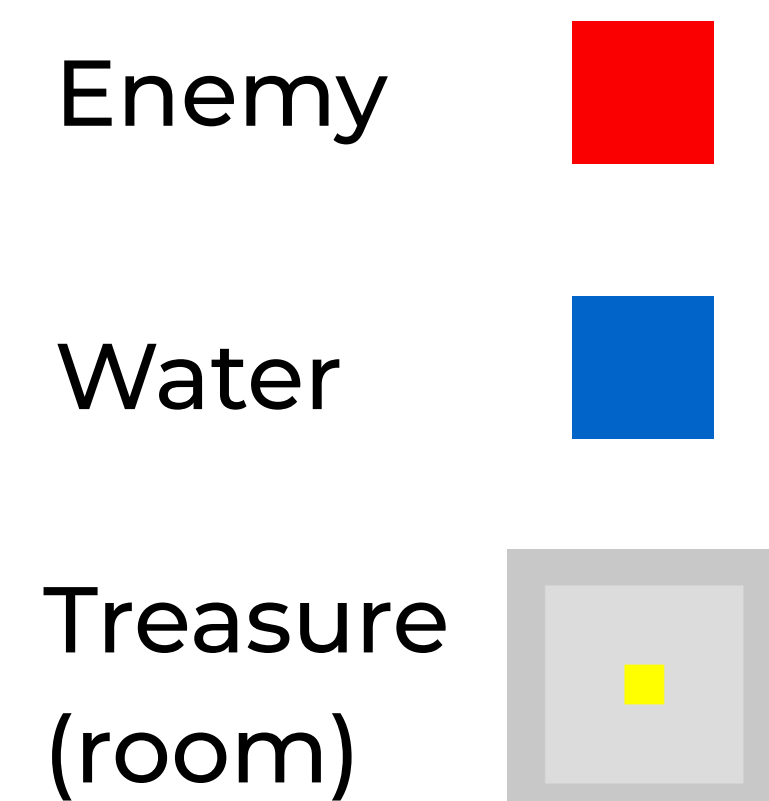
Methods

- Four Biomes are created
- Each have thier own unique cellular automata parameters and rules for cave generation and water generation.
- Layered Automata: Water is generated with another CA overlapping the previous caves

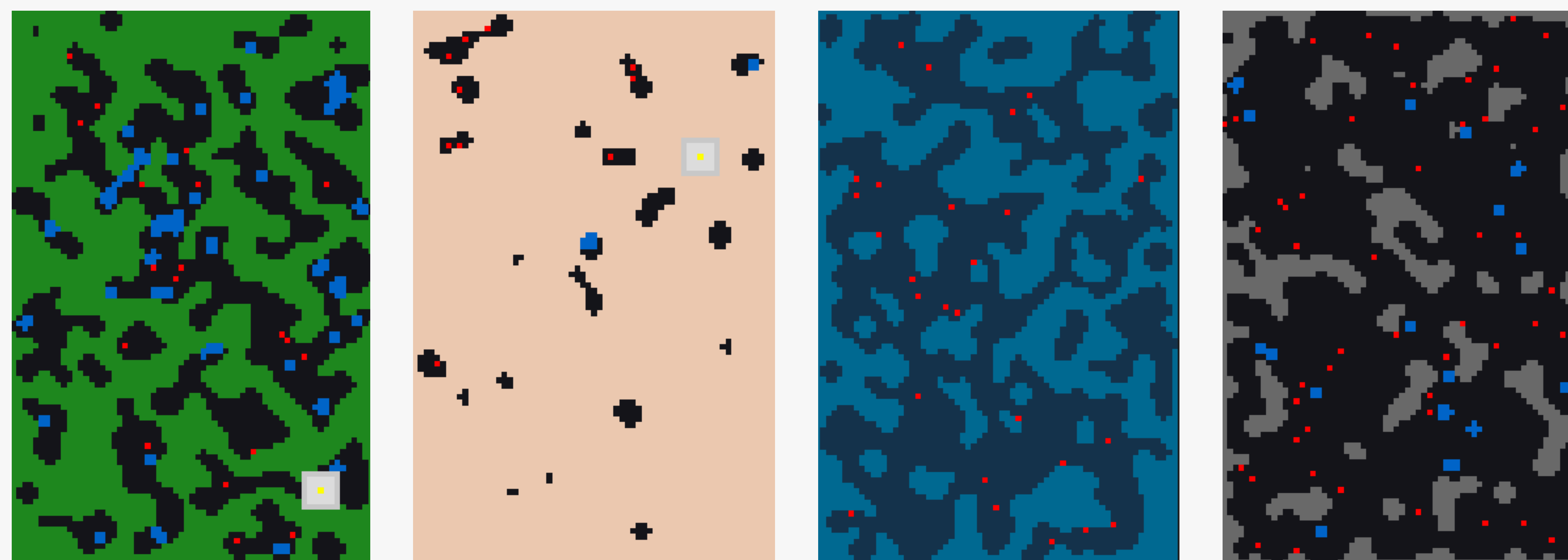


Biome/ Parameters	Alive %	Neighbor Threshold	Enemy Spawn (per block)	Water %	Treasure (per biome chunk)
Forest	53	3+ Dead→ Dead 4+ Alive → Alive	1%	30	40%
Desert	65	5+ Dead→ Dead 4+ Alive → Alive	5%	30	100%
Underwater	50	4+ Dead→ Dead 4+ Alive → Alive	1%	100	20%
Mountain	40	2+ Dead→ Dead 4+ Alive → Alive	1%	20	20%

Key



Results



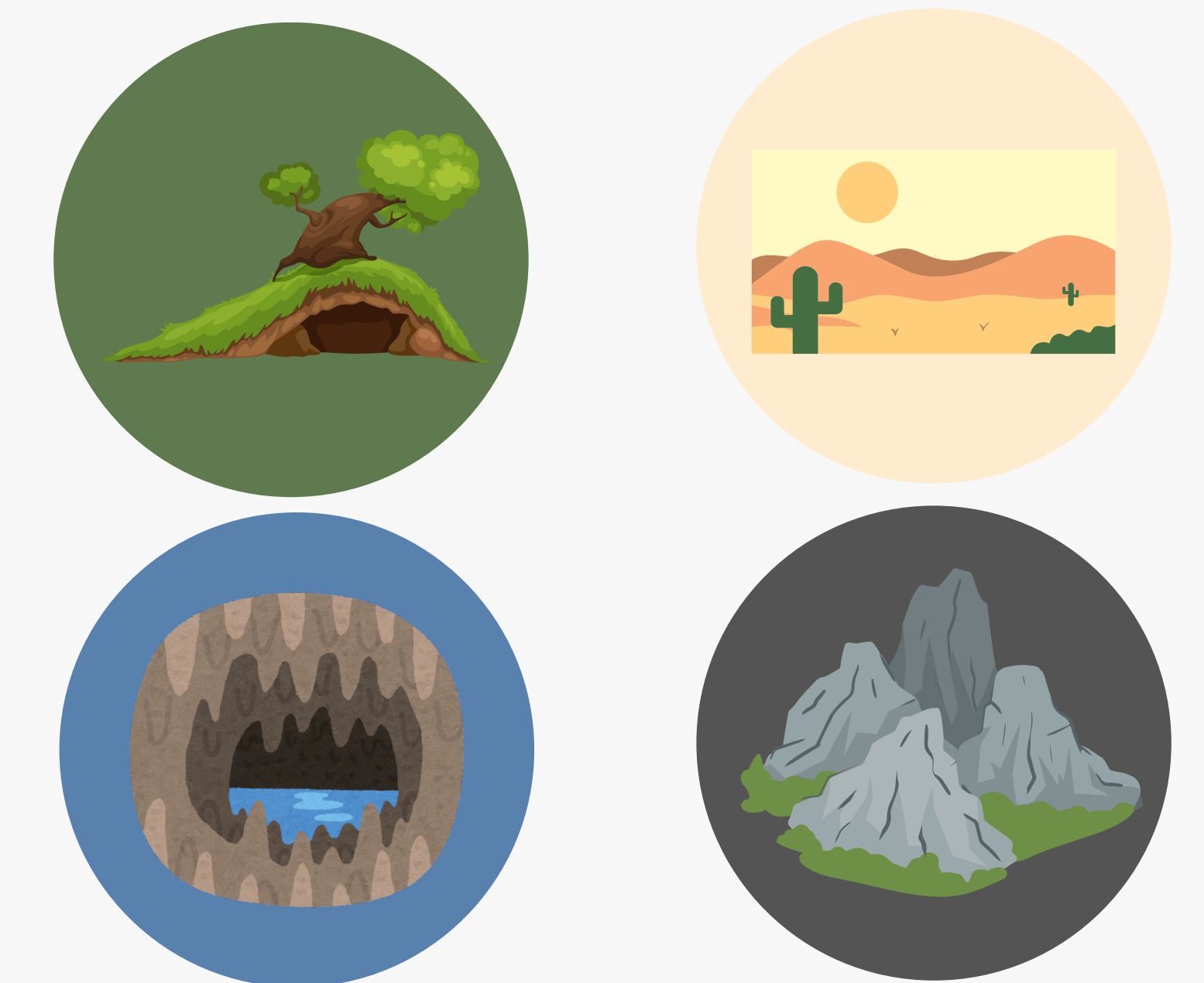
Forest

Desert

Underwater

Mountain

- **Forest:** Spawling caves, good chance of water and treasure
- **Desert:** Small patches of cave, less water, 100% chance of treasure
- **Underwatcer:** Larger, water filled caves. Low chance of treasure.
- **Mountain:** Large caverns, more enemies, low chance of treasure



Conclusion

- Unique cellular automata rules and parameters can lead to interesting varieties of cave generation in video games/simulations.
- These unqiue cave systems keeping games interesting and enganging to players